

Aoyu Liu

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B.S. in Mathematics and Computer Science Yingcai Honors College UESTC Image Processing Lab

PROFILE

I am Aoyu Liu, an undergraduate student at the Yingcai Honors College, University of Electronic Science and Technology of China, where I am pursuing a B.S. in Mathematics and Computer Science.

By the time of my sophomore year, my research focused on low-level computer vision. I work under the guidance of [Prof. Shuaicheng Liu](#) in the Image Processing Lab at UESTC. During this period, I am fortunate to work closely with [Dr. Zhen Liu](#).

I also have practical experience in research computing operations and currently serve as the administrator of two 8-GPU RTX 4090 servers. My research interests are now expanding from low-level computer vision toward embodied intelligence, with a particular focus on manipulation and vision-language-action models.

Low-level Computer Vision

Embodied Intelligence

Manipulation

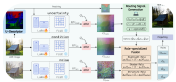
Vision-Language-Action Models

HDR Reconstruction

NEWS

- **2026:** Our work *Unleashing Pretrained Generative Priors for Single-Image HDR Reconstruction via Uncertainty-Routed Attention* is in submission to **SIGGRAPH Asia / ACM TOG**.
- **2026:** Our paper *ExpoCM: Exposure-Aware One-Step Generative Single-Image HDR Reconstruction* is accepted to **CVPR 2026**.
- **2025:** I received the University Special Scholarship from the University of Electronic Science and Technology of China.

SELECTED PUBLICATIONS



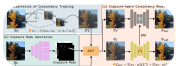
Unleashing Pretrained Generative Priors for Single-Image HDR Reconstruction via Uncertainty-Routed Attention

Aoyu Liu et al.

ACM Transactions on Graphics (TOG), SIGGRAPH Asia, in submission

Research on pretrained generative priors for single-image HDR reconstruction with uncertainty-routed attention.

[Exhibition](#)



ExpoCM: Exposure-Aware One-Step Generative Single-Image HDR Reconstruction

Aoyu Liu, Zhen Liu, Ziyi Wang, Dian Chen, Bing Zeng, Shuaicheng Liu

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

Research on exposure-aware one-step generative reconstruction for single-image HDR imaging.

[arXiv](#) [Exhibition](#)

COLLABORATORS

Zhen Liu

Ziyi Wang

Shuaicheng Liu